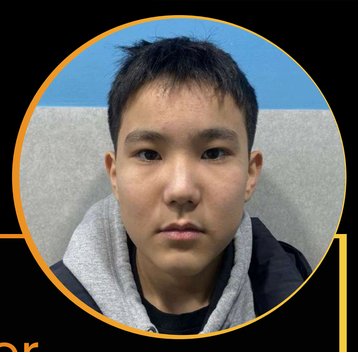
PORTFOLIO
OFF SEASON

SPIRIT
#2544

SPIRIT
#25547

TEAM STRUCTURE

We are a team from the Schoolchildren's Palace named after Al Farabi in Astana. Our team consists of 8 members from different schools.



Diar
Inspire manager
Freshman
Helps with CAD and developing outreach projects.



Ernar
Lead Coder
Freshman
Writes code and programs autonomous routines.

MENTORS



Mansur
Helps with 3D modeling, building, and printing while teaching our team skills, techniques, and problem-solving.

MAIN GOAL

Our main goal for the season was to win an Connect award at the CAFC 26 and qualify for the World Championship.



Khamza
Lead Engineer
Junior
Builds the robot and works on its mechanisms.



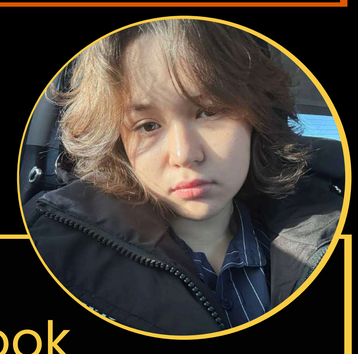
Nurdaulet
Coder
Freshman
Helps with coding, autonomous routines, and other programming tasks.



Inkarov Bekzat
Helps us with CAD and our robot, teaching the team design skills and problem-solving techniques.



Freshman
Viktoria
Outreach manager
Organizes outreach, forums, hackathons, and more.



Freshman
Darina
Handbook engenner
Creates the portfolio, engineering notebook, and design.



Freshman
Aruna
SSM
Manages social media, creates content and engages the community.

OUR JOURNEY

After FGC, the Munara team disbanded, and one former member, Khamza, founded a new team—Spirit. It is now in its second generation.

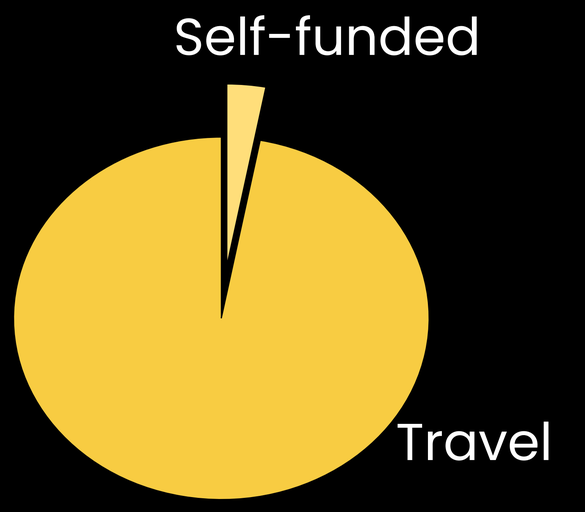
FINANCE

Okushylar Sarayi has supported our team since its founding. This season, we reached over 100 companies with our sponsorship package and continue seeking more sponsors.





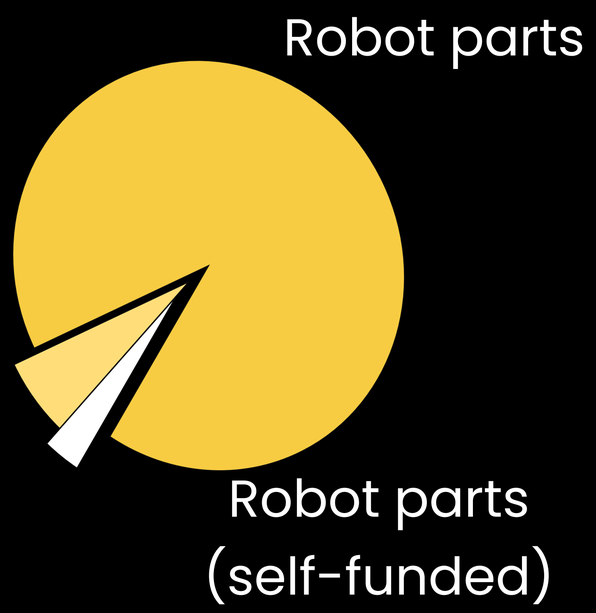
INCOME



RISK AND SOLUTION

risk	solutions	outcome
Lack of funding	Active sponsor search, send over 100 companies our sponsorship	Active sponsor search resulted in continued team operation this season.
Shortage of robot parts	Requesting new parts from spons	The sponsor was buying the necessary parts
Team cohesion challenged by members from different schools	Team-building, lab collaboration, and cross-subteam tasks	Stronger relationships, better communication, and improved team morale

EXPENSES



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SOCIAL MEDIA



Our Instagram account is our main social media, which we use to promote FIRST.

VOLUNTEERING & ORGANIZING

1000+ PARTICIPANTS IN 2 YEARS



From 2023 to 2025, our team volunteered at the international RoboZerde robotics competition

FIRST KAZAKHSTAN

NEWS FTC



We took over the news FTC account for New Zealand together with #30060.

IMPACT

- **8** MEMBERS
- **+600** NEW PEOPLE INTO FIRST
- **2 NEW** FIRST TEAMS STARTED
- **53K** VIEWS IN INSTAGRAM
- HAD MEET WITH **60+** TEAMS
- **22** EVENTS WE HOSTED

PLAN FOR THIS SEASON AND OFF SEASON

- Completed the main parts of our robot and conducted various outreach events.
- Worked on outreach and promoted STEM across Astana.
- Host Bilim Shyny off-season event 2026.
- Do team building to strengthen the team.

PLAN FOR THE FUTURE

- Became more familiar with the next season while developing our outreach and engineering.
- Rested and prepared for the upcoming 2026–2027 FTC season.

TIMELINE

- 07/01/25 — Online conference for FLL
- 11/01/25 — FARABI CUP
- 12/01/25 — Internatiomal conference wirh #19176 and #3747
- 17/01/25 — SPIRIT-pen Sai
- 04/02/25 — Charity Fair
- 06/02/25 — Online conference
- 10/02/25 — Robotics: spirit in motion
- 12/02/25 — The best project
- 14/02/25 — Online conference with #13617 and #8721
- 15/02/25 — Tech forum with #22975 and #21059
- 16-17/02/25 — Hackathon
- 12-13/04/25 — BILIM SHYNY 2025
- 19/04/25 — Master class with participant of the Asian championship of WCA "Rubik"
- 17/12/25 — News FTC new Zealand with #30060
- 02/01/26 — The international forum with #19654 and #14380
- 10/01/26 — FIRST Forum with STEM community
- 01/02/26 — Free robotics workshop
- 04/02/26 — Award and Portfolio workshop wirh #28659

SPIRIT #25547

SEASON HIGHLIGHTS

- BFR 25 — Reach award winner
- AR 25 — Reach award 2nd
- AFQ 26 — Connect award 3nd
- CAFC 26 — 21nd place. 5 win \ 3 lose

HOW WE DEVELOPED OUR SKILLS



Our builder took 3D modeling course to improve CAD skills.



Our designer attended EI Umiti Camp 2025, improving video editing and ranking top.

DEVELOPMENT OF NEW MEMBERS

Our outreach subteam rotates tasks, letting members gain hands-on leadership experience. For example, Viktoria hosted a FIRST Forum as soon as she joined the team.



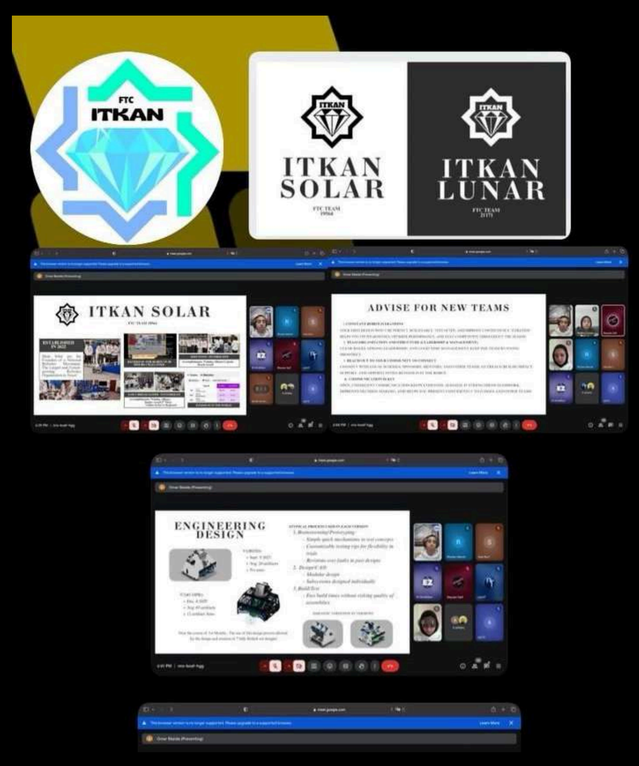
EVENTS

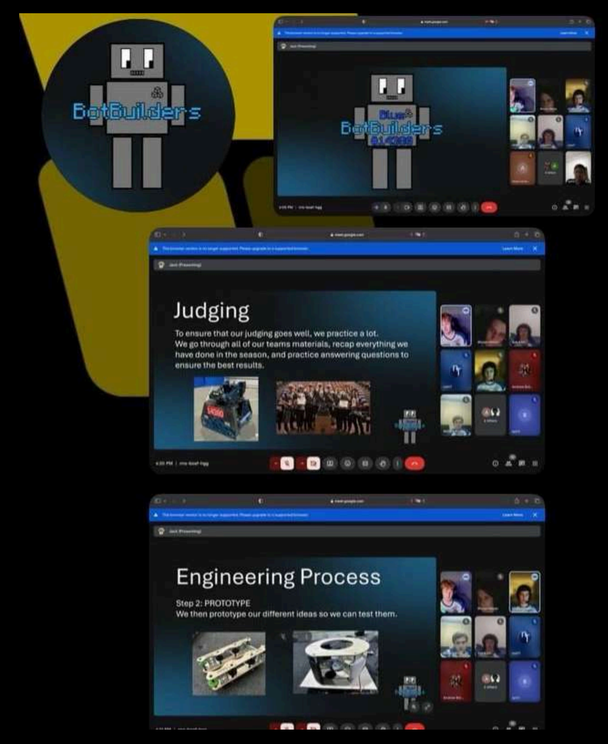
INTERNATIONAL FORUM

REACHED 15 PEOPLE

On January 2nd, our team hosted an international forum with #19564 and #14380

Two international teams—BotBuilders #14380 itkan Lunar #21171 and Itkan Solar #19564—shared their journey, engineering process, and outreach experiences. We are very grateful to them!





FIRST FORUM









FIRST FORUM

REACHED 15 PEOPLE

On January 10, our team and FarabiRobotics #310112 hosted a forum, inviting STEM community members Uali Zeinur, Asylbek Myrzhahmet, Kuanysh Magauin, and Yevangelina.

The forum covered FLL and FTC topics, with speakers sharing team experiences, challenges, mentoring tips, judging insights, leadership advice for captains, and guidance on finding sponsors.

SPIRIT PEN Şäl

We created “Spirit Pen Şäi” to share stories; new members featured Teams AMD #28588, TOLQYN #33595, and FIZMAT robotics #28061





AWARDS AND PORTFOLIO WORKSHOP

REACHED 10 PEOPLE

We with Team Hotweels #28659, we hosted an online workshop where teams shared portfolio tips and award strategies.



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ONLINE CONFERENCE FOR FLL TEAMS





Registration link in post description.
See you there!

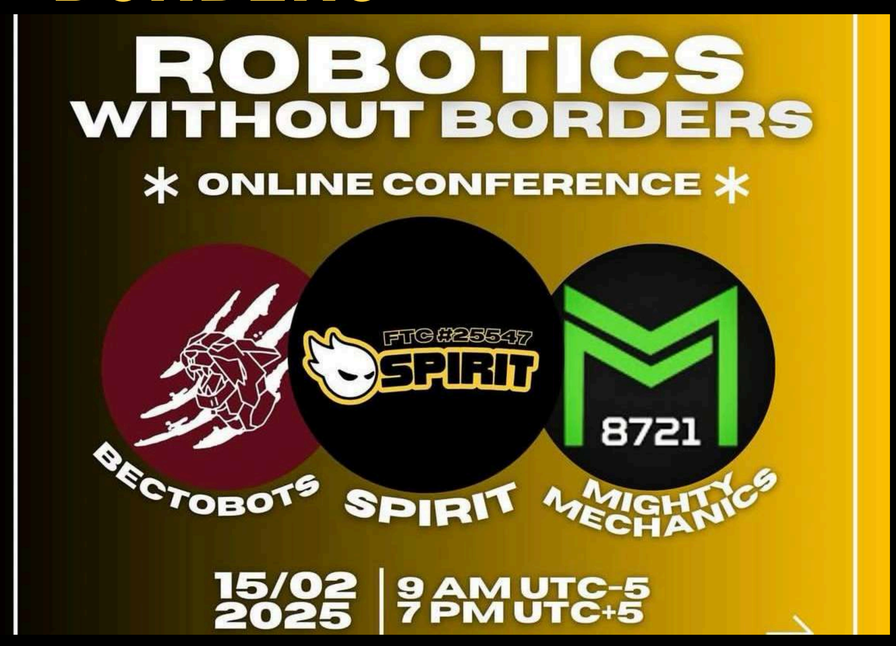
Two invited speakers assisted teams by sharing professional advice

TECH-SPIRIT FORUM



We hosted Tech-spirit forum with Teams Bilorda #22975 and Qtech #21059

ROBOTICS WITHOUT BORDERS



We hosted the FTC online conference “Robotics Without Borders” with team presentations

FARABI CUP



We hosted the FARABI CUP robotics competition at Palace of Schoolchildren

INTERNATIONAL CONFERENCE



We hosted an international conference, collaborating with Brobotic #19176 and Hive #3747

SPIRIT PEN SAI



We provided published resources by sharing two interview episodes with Teams Bilorda #28313 and Venom #24783

HACKATHON



We hosted an online sustainability hackathon focused on solutions for ocean-related environmental issues.

CHARITY FAIR



We hosted a Charity Fair to raise funds and support children from orphanages.

CONFERENCE WITH ZERTTE



We hosted an online conference with Team Zertte, where teams shared knowledge

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ROBOTICS: SPIRIT IN MOTION



We hosted an event with a Robot Reveal, team experience, and an interactive activity with prizes.

THE BEST PROJECT



We hosted the Forum, where teams competed for the “Best Project” award of 30,000 KZT.

ROBOTICS WITHOUT BORDERS



We hosted a speedcubing workshop and mini-tournament with a WCA Asia 2022 champion.

FREE ROBOTICS WORKSHOP



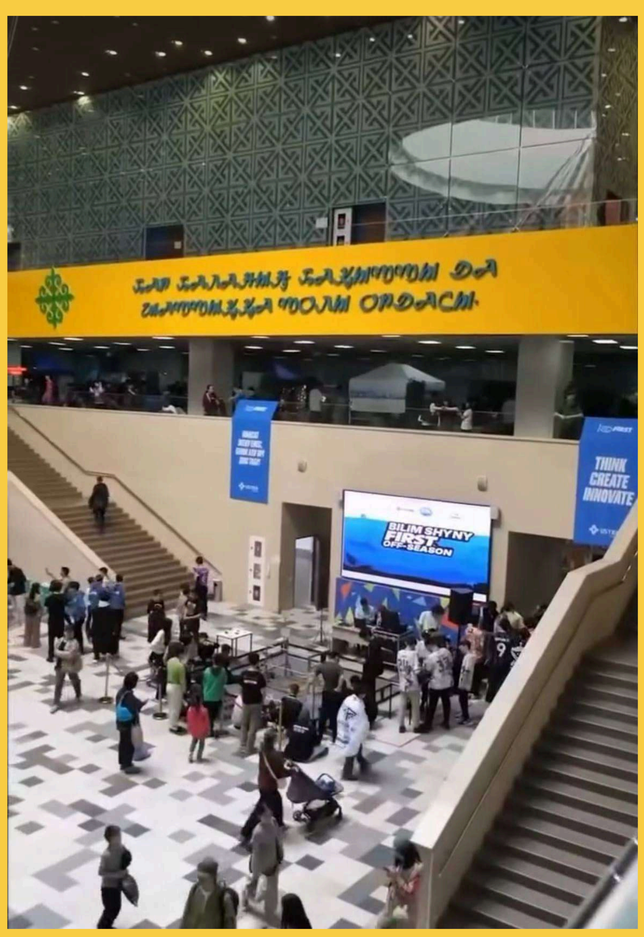
We hosted and mentored a two-day FLL and FTC session with workshops, practice, and gameplay.

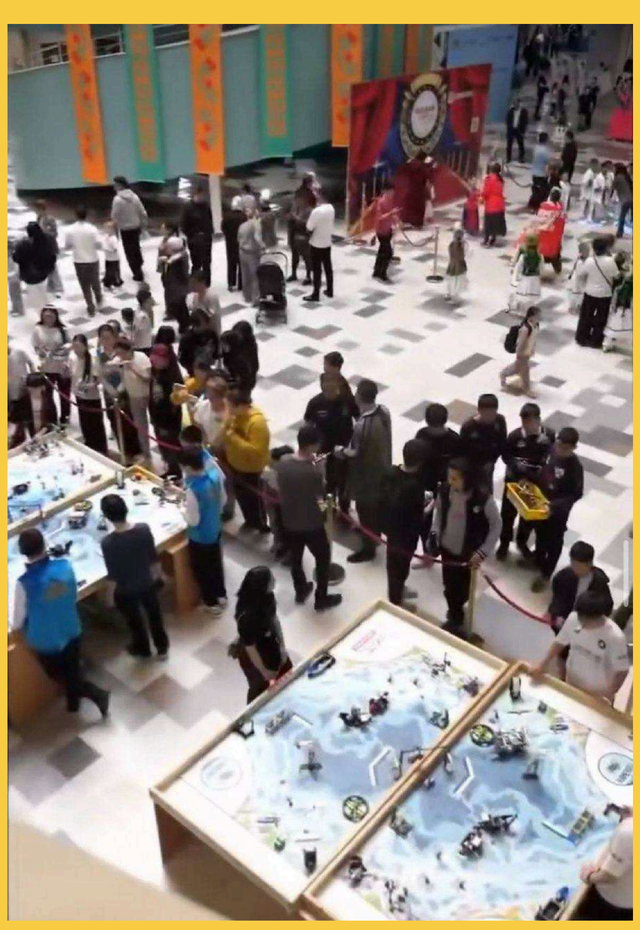
BRIDGE TO TECH



We are launching SPIRIT X MEGADETH FLL team From blocks to code, from ideas to metal — featuring Hub, Evolution, and Engineering Journal
Featuring over 2,100 REV and GoBILDA models to make CAD work faster and more efficient.

OFF SEASON. BILIM SHYNY





REACHED 600 PEOPLE

We hosted the BILIM SHYNY off season. That was held in Astana on April 12–13. The event brought together over 610 students from 8 regions of Kazakhstan and featured all FIRST categories. We plan to host a BILIM SHYNY this year

SPIRIT
#25547

STEM SPREADING



No 51 school. Reached 60 people
20/09/25
We presented FTC, our robot, and AI in engineering, inspiring students to explore STEM.



BILIM PARK. Reached 20 people
30/04/25
We hosted an FLL masterclass at BILIMPARK, teaching robotics, teamwork, and hands-on problem-solving.



No 5 school. Reached 60 people
11/09/25
We hosted a masterclass, demonstrating our robot and sharing engineering and teamwork tips



No 103 school. Reached 45 people
06/09/25
We hosted a robotics masterclass at School-Lyceum No. 103, teaching coding and engineering, and helped form two new FTC teams

MENTORSHIP

We started two FTC teams — Oreo103 and Frozen MurMur at School No.103 last season, supporting onboarding and team setup.









We mentored Farabi Robotics helped them achieve a 540-point robo-game high score at Central Asia 2026.



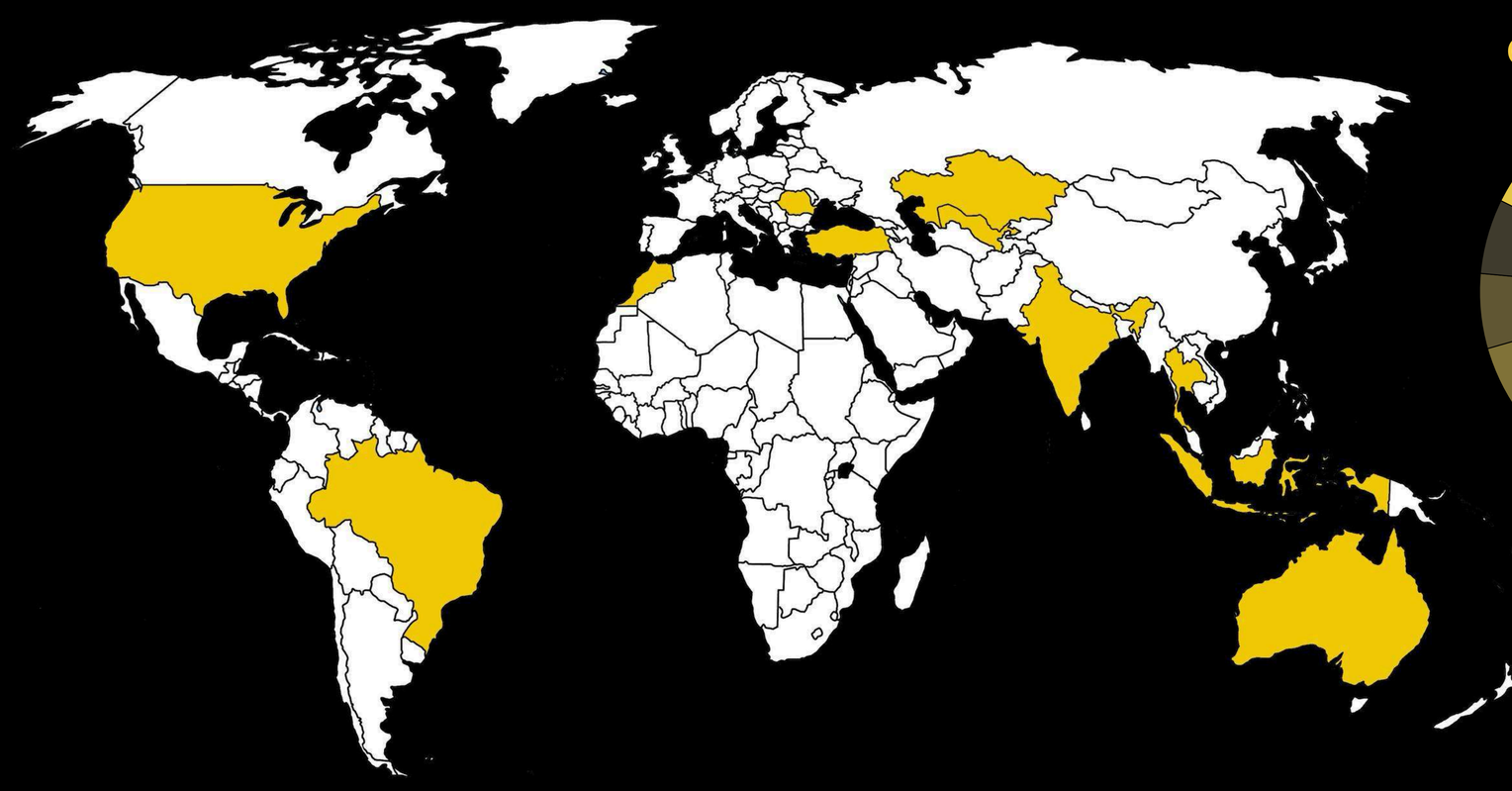
OUR MENTORSHIP. WHAT WE LEARNED?

First Impression: Mentored Farabi FLL and SnapMakers
Mentorship & Support: Guided teams with structured technical and team support, helping them complete their FIRST projects.
What We Learned: Improved leadership, communication, and ability to explain ideas clearly.

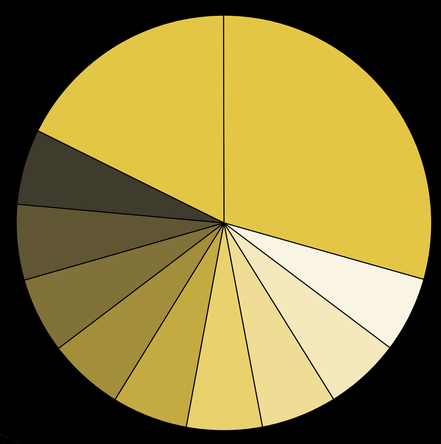
Won the Motivate Award at the Astana Regional Championship and hosted an Arduino workshop recognized by judges in 2025.

SPIRIT
#25547

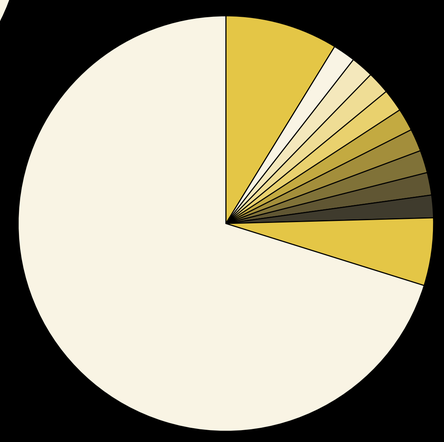
CONNECT



We held a meeting with about 60 teams, including one-third international,



international



in the world

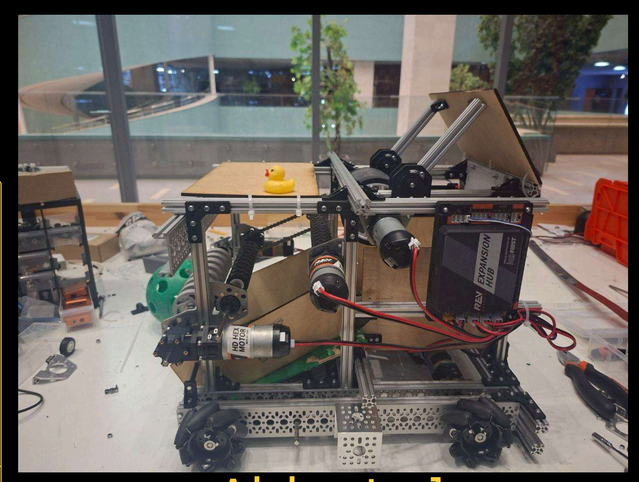
USA - 5, Brazil - 1, Morocco - 1, Turkey - 1, India - 1, Thailand - 1, Australia - 1, Kyrgystan - 1, Indonesia - 1, Romania - 3, Kazakhstan - 40

ENGINNERING PROCESS
BRAINSTORM

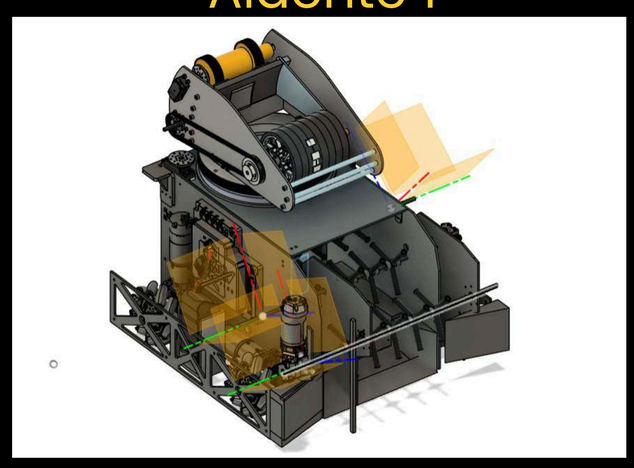
THE IDEA **TO SOLVE** **CAD** **BUILDING** **TEST** **DATA LOG**

At the DECODE season start, "Aldente" focused on reliability, fast repairs, high-volume scoring, and optimized performance.

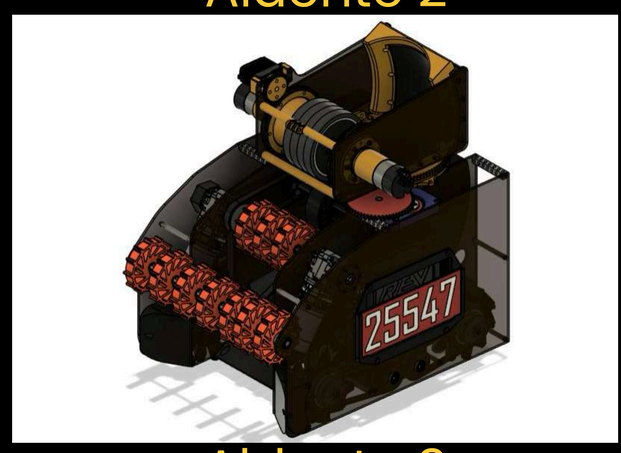
the idea	**sketch**	**cad**	build	test	data log
We evaluated scoring methods and chose a vertical lift over an angled shooter based on strategy.	Initial scale drawings defined the footprint and ensured compliance with the 18-inch size limit.	Used to check interference and simulate stress on moving parts.	Transitioned from prototype materials (wood) to final materials (aluminum and PETG 3D prints).	Each mechanism underwent 50-cycle stress tests to check for heat buildup and loose hardware.	All test results were logged in our engineering records and guided the next design iteration.



Aldente 1



Aldente 2



Aldente 3

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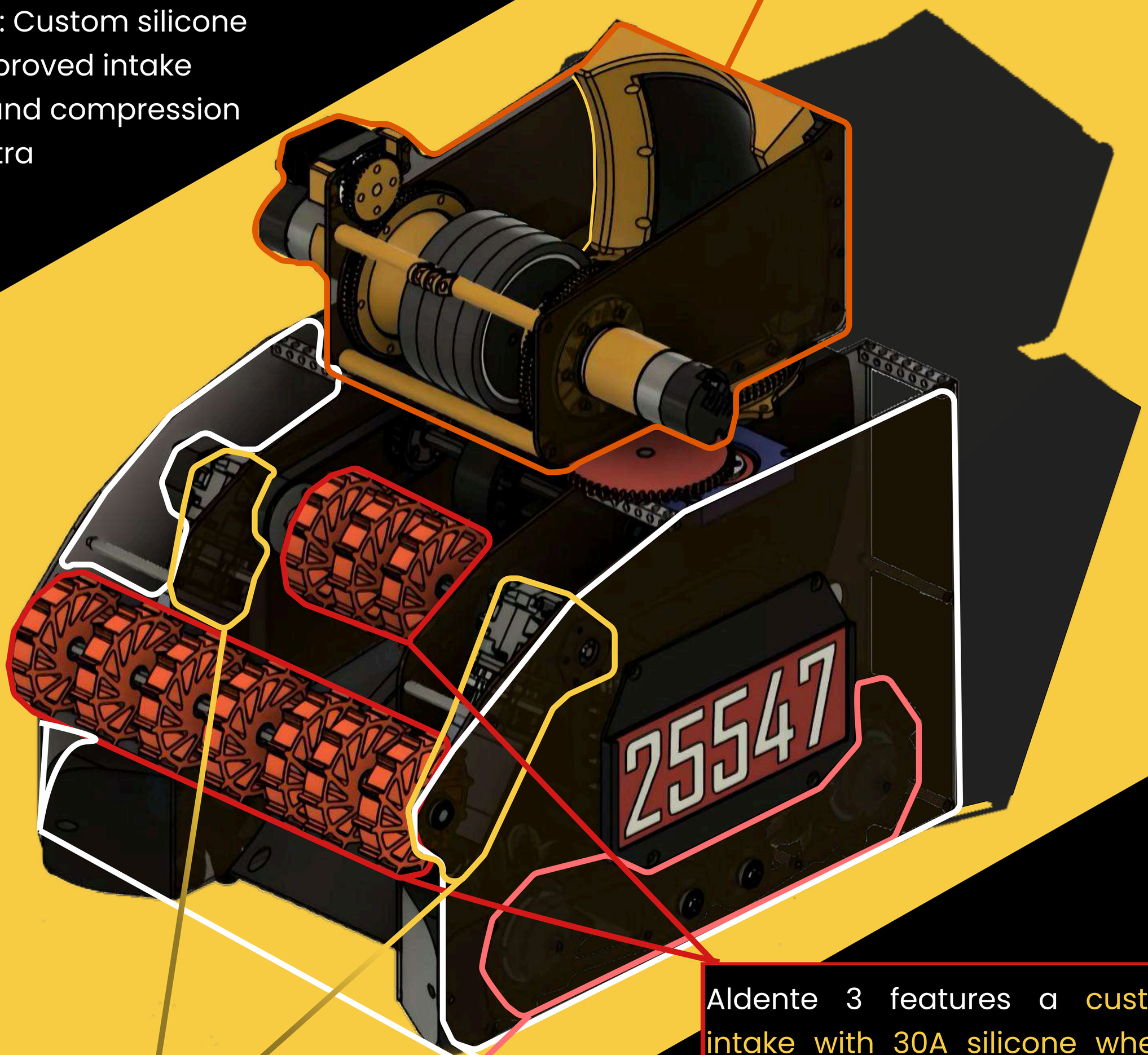
ALDENTE 3 FINAL VER

Components: Two 6000 RPM motors, adjustable hood, servo-controlled direction/ gate, rotating turret, vision camera.
Upgrades: Wider transfer wheels, updated gears, consistent ball compression.
Strategy: Optimized flywheel speed for scoring.

We added a 3D-printed, manually adjustable hood for easier and more precise shooting.

Innovation: Custom silicone wheels improved intake reliability and compression without extra motor load.

Strategy: Intake designed for fast, consistent artifact collection.



Aldente 3 features a custom intake with 30A silicone wheels and surgical tubing for smooth, reliable artifact movement.

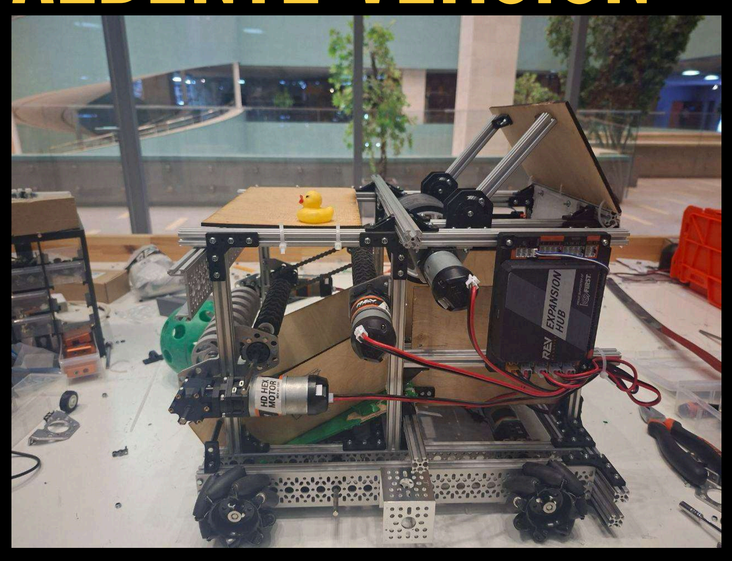
Two intake motors—one per side for balance—delivering 2400 RPM and 10 kg·cm combined torque.

The robot uses a 435 RPM belt-driven drivetrain, balancing speed and pushing power for fast, controlled scoring.

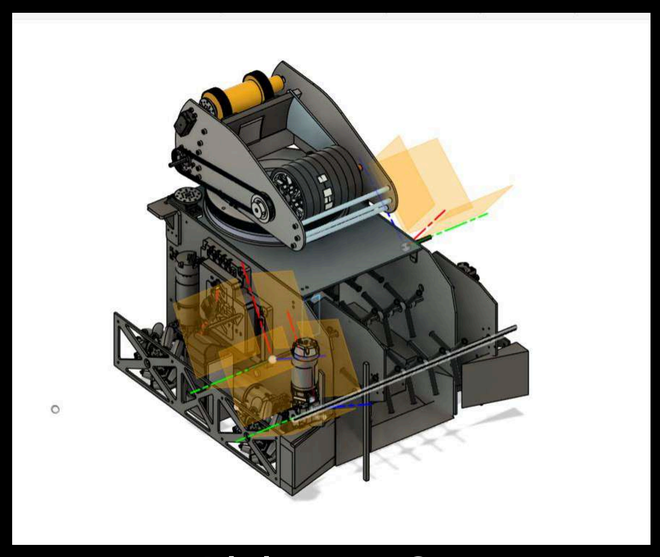
To reduce weight and cost, we replaced aluminum parts with engineered plywood, lowering mass while keeping strength and improving repairability.

SPIRIT
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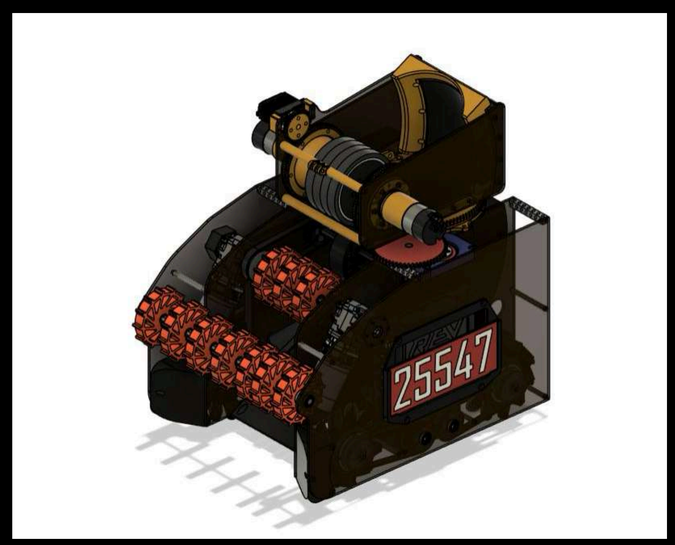
ALDENTE VERSION



Aldente 1



Aldente 2



Aldente 3

ALDENTE 1

Aldente 1 — Performance Challenges

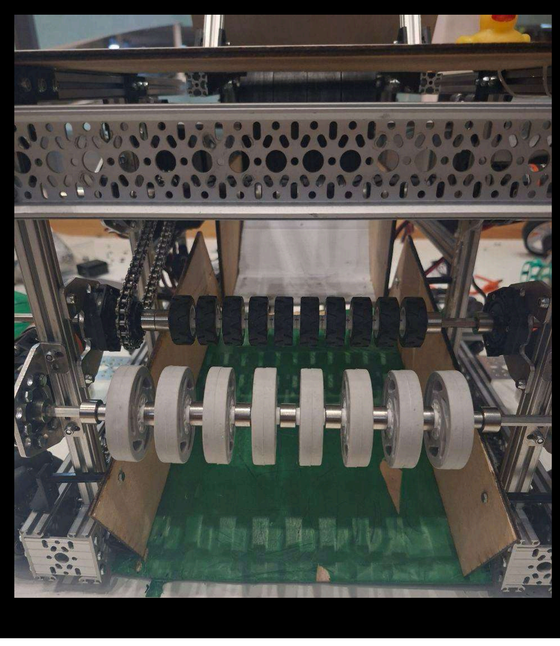
Our first robot, Aldente 1, revealed key weaknesses early in the DECODE season:

High weight (13 kg) causing slow movement

Limited to one artifact per cycle

Low shooting accuracy (1/10 shots) due to tight flywheel spacing



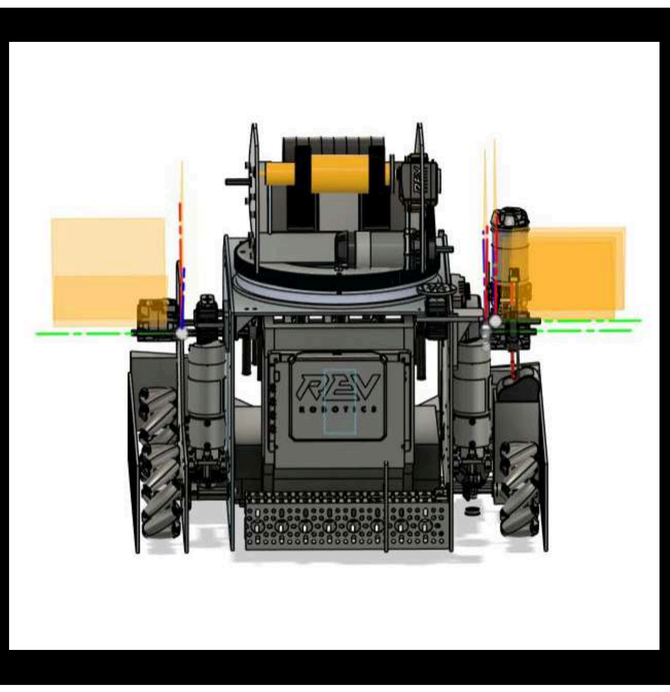


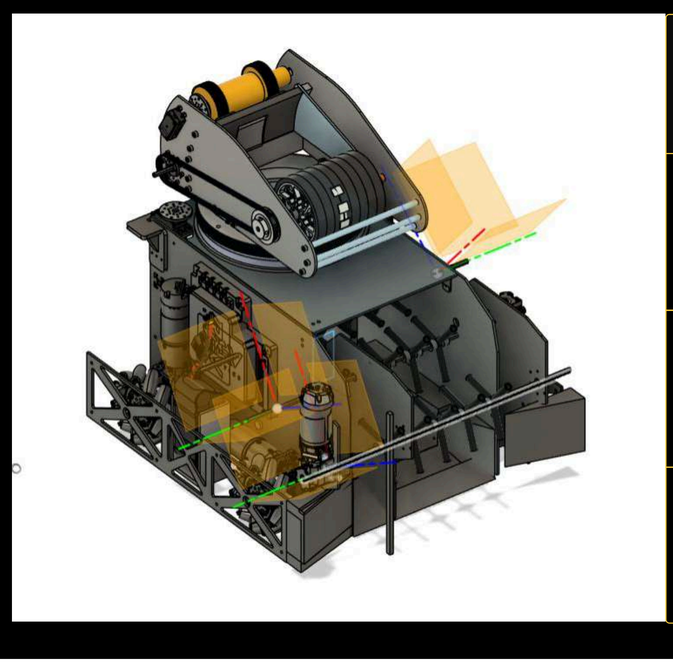
ALDENTE 1.5

Aldente 1.5 added a second shooter motor, integrated intake, and increased flywheel spacing, improving performance but requiring a full redesign for competitive cycles and capacity.

ALDENTE 2

Aldente 2 was fully custom, with a parallel plate drivetrain, rotating shooter, and redesigned layout for three-artifact capacity and cycle efficiency.





Advantages	Trade-offs and Limitations
capacity increased to three artifacts	significant loss of internal space due to suboptimal layout
faster aiming due to the rotating shooter	incorrect geometry in several areas, complicating maintenance
improved scoring potential compared to earlier versions	increased overall weight caused by the aluminum frame

CHALLENGE 1 — CYCLE TIME OPTIMIZATION

Problem: ~20s cycle due to high weight, slow intake/transfer, and sequential shooting.

Analysis: System-level breakdown (movement, intake, transfer, shooting) showed full redesign was needed.

Solutions: Reduced weight (plywood), redesigned intake/transfer for 3 artifacts, implemented burst-fire shooting.

Result: Faster cycle time without increasing complexity.

Weight reduction	Intake and Transfer	Shooting vethod
improved acceleration	reduced unnecessary driving	3 artifacts are fired in 500 ms
reduced field traversal time	fewer stops and restarts	minimized time spent at the goal
lower drivetrain load	increased cycle efficiency	increased scoring predictability

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CHALLENGE 2 — INTAKE MOTOR OVERHEATING

Problem: Intake motors overheated during extended use.

Analysis: High load from inefficient gear ratio caused overheating and performance loss.

Takeaway: We identified issues through testing, applied analysis, and implemented data-driven solutions.

Solution	Result
A smaller intake pulley improved responsiveness.	Motor temperatures decreased.
A larger transfer pulley prioritized torque over speed.	More stable operation and improved reliability.

CHALLENGE 3 — TRANSMISSION EFFICIENCY AND OVERHEATING (CORRECTED VERSION)

Problem: Gear-driven transmission caused overheating, energy loss, and low efficiency at high speeds.

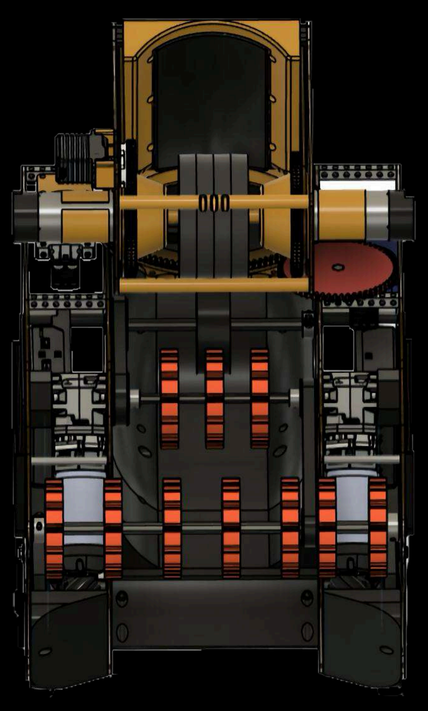
Analysis: High friction in gears reduced performance and reliability during fast operation.

Lessons Learned: We learned to treat the robot as an integrated system, balancing weight, ratios, and design based on real testing.

Solution: Replaced gear transmission with a belt-driven system for smoother, more efficient power transfer.

Result: Reduced heat, improved efficiency, and significantly increased speed without added complexity.

INTAKE SYSTEM — FRONT VIEW



Early intake designs were too rigid, causing overheating and wear. Testing led to a custom intake better suited to the robot.

Engineering Changes	Result
redesigned intake geometry	the intake system unlocked the robot's full potential
Used silicone wheels with controlled compression.	ball firing speed increased by 300%
Enhanced integration with the transfer system.	Motor load and overheating were greatly reduced.

Two-stage transfer system ensures stable artifact movement. Custom pulleys with optimized gear ratios boost output speed to 3600 RPM.

SHOOTER SYSTEM



Main Components:

Two 6000 RPM shooter motors
Adjustable hood
Servo-controlled ball direction and gate
Rotating turret with servo

Additional Changes:

Redesigned transfer system with wider wheels
Updated gear ratios
Consistent compression at every stage

Game Strategy: Optimized flywheel speed for scoring.

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PROGRAMMING

SENSORS — AUTOMATION TO IMPROVE MECHANISM PERFORMANCE

goBILDA Mechanical & Control Integration

Hard Stops: Repeatable zero position (±1–2 mm) for accurate cycles.

Analog Encoders: <1° resolution for stable, repeatable positioning.

Odometry: Dead wheels + IMU give ±1–2 cm position accuracy, even with wheel slip.





goBILDA 4-Bar + Pinpoint Odometry

High-precision odometry with IMU for repeatable autonomous routines

~1500 Hz pose updates for smooth, high-frequency control

Mecanum kinematics improve path accuracy and reduce endpoint error

PEDRO PATHING AUTONOMOUS MOTION PLANNING & CONTROL

Spline Pathing

Uses spline curves (e.g., Bézier) for smooth, continuous paths

Eliminates abrupt velocity changes, reducing mechanical stress and wheel slip

FTC PID Pedro Pathing Tuning

Custom PID controllers minimize steady-state error and oscillation

Reduces trajectory deviation by 30–45%

Continuously corrects pose to limit accumulated error during long autonomous routines



Integration in Robotics

Ensures reliable, repeatable autonomous scoring with balanced speed and accuracy

MATCH STRATEGY

Autonomous and Tele-Operated Performance

Autonomous Strategy:

Executes a 6+0 scoring plan, prioritizing consistency

Cycle time: 2.5–3 s per element (alignment, intake, placement)

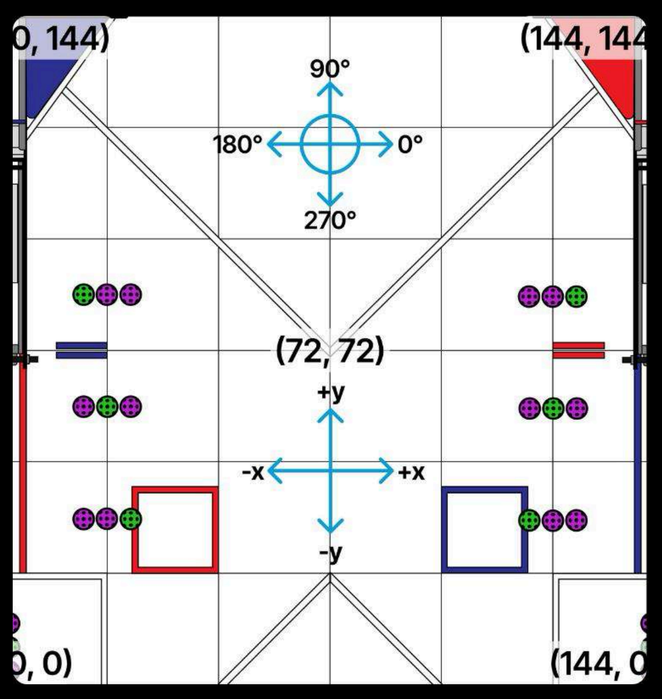
Pathing accuracy: ±2 cm and ±2° final pose

Success rate: 85–90% over repeated runs

Total routine: 12–14 s, leaving margin for field variability

Strategic Impact: Stable drivetrain improves consistency and ensures reliable, sustained performance.

PATHFORGE — VISUAL AUTONOMOUS PLANNING & DIAGNOSTICS TOOL



Pathforge is a visual, data-driven platform that lets FTC teams design and test autonomous paths before the robot is built, speeding up development.

SPIRIT #25547

CONTROLS

Left Strick	Right Strick	LB	RB	Y	X
DRIVE/ STRAFE	TURN/ ROTATE (ROBOT HEADING CONTROL).	INTAKE IN (COLLECT).	INTAKE OUT / REVERSE (EJECT).	LOSE BLOCKER (LOCK).	OPEN BLOCKER (RELEASE).

Pathforge lets teams visually design structured autonomous paths with consistent simulation and robot execution. Built-in diagnostics track errors for data-driven tuning, making it easy for beginners and efficient for advanced teams.

Pathforge Components:

Pathforge Studio: Web-based visual tool to design and preview FTC paths

Pathforge Core: Lightweight Java library that runs the same paths on the robot